ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

ROBOTICS AND INDUSTRIAL AUTOMATION

PAPER 2 · PRACTICE B · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 4 Robotics and Industrial Automation paper 2 · practice b.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers, sensible design choices and neat, recognisable diagrams.

SECTION A

QUESTION 1

1.1 1.1.1 A ✓ 1.1.2 B ✓ 1.1.3 A ✓ 1.1.4 D ✓ 1.1.5 A ✓ 1.1.6 B ✓ 1.1.7 C ✓ 1.1.8 D ✓ 1.1.9 B ✓ 1.1.10 C ✓ (10)

1.2 1.2.1 True ✓ 1.2.2 True ✓ 1.2.3 False ✓ 1.2.4 True ✓ 1.2.5 True ✓ 1.2.6 False ✓ 1.2.7 True ✓ 1.2.8 False ✓ 1.2.9 False ✓ 1.2.10 True ✓ (10)

1.3 1.3.1 C ✓ 1.3.2 B ✓ 1.3.3 G ✓ 1.3.4 I ✓ 1.3.5 E ✓ 1.3.6 J ✓ 1.3.7 F ✓ 1.3.8 D ✓ 1.3.9 A ✓ 1.3.10 H ✓ (10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2

2.1 Robot in the centre ✓; infeed conveyor ending in the robot's reach ✓; two pallet stations within reach ✓; fence around the cell with an interlocked access gate ✓; light curtain(s) at the pallet entry / exit openings ✓; control cabinet and HMI outside the fence, neat and labelled ✓ (6)

2.2 $\text{Payload} = 8 + 4 = 12\ \text{kg}$ ✓; choose the 20 kg robot (10 kg is too small) ✓; other specification, any ONE ✓: reach to all pallet corners and the top layer; speed / cycle time; number of axes (4-axis palletiser); repeatability. (3)

2.3 $\text{Boxes per pallet} = 5 \times 6 = 30$; $\text{time} = 30 \times 6 = 180\ \text{s}\ (3\ \text{min})$ ✓✓; $\text{boxes per hour} = 3\,600 \div 6 = 600$ ✓; $\text{pallets per hour} = 600 \div 30 = 20$ ✓ (4)

2.4 While the forklift removes a full pallet and puts down an empty one at one station ✓, the robot keeps stacking at the other station ✓, so production does not stop and the output rises ✓. (3)

2.5 Vacuum (suction) gripper ✓ – grips the flat top of sealed boxes quickly without damaging them (a fork / clamp gripper accepted with a sound reason) ✓ (2)

[18]

QUESTION 3

3.1 Any FIVE sensible inputs with addresses ✓✓✓✓✓, e.g. START, STOP, emergency stop OK, box at pick position, pallet present station A, pallet present station B, light curtain OK, gate closed, robot ready. Any THREE outputs ✓✓✓, e.g. infeed conveyor motor, robot start / pick signal, vacuum valve, pallet-full lamp, alarm beacon. (8)

3.2 START with a seal-in contact in parallel ✓; in series with NC STOP ✓, E-stop contact ✓ and NC box-at-pick sensor ✓; coil Conveyor ✓, e.g. (5)

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|--+--[ START ]--+--[/ STOP ]--[ E-STOP OK ]--[/ BOX AT PICK ]--( CONVEYOR )--|
|  |             |
|  +--[ CONV ]---+
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3.3 Each "box placed" signal from the robot increments a CTU counter with preset 30 ✓; when the done bit is set the PLC signals "pallet full", lights the lamp and tells the robot to use station B ✓✓; the counter is reset when the full pallet is removed / the new pallet is detected ✓. (4)

3.4 Signals passed in both directions so that each controller knows what the other is doing and they stay in step ✓✓, e.g. PLC sends "box ready" and the robot answers "box picked", then the PLC moves the next box / robot sends "pallet complete" and the PLC replies "new pallet ready" ✓. (3)

[20]

QUESTION 4

4.1 Any FOUR ✓✓✓✓: running / stopped / fault status; boxes on the current pallet and pallets completed; active alarms with messages; which pallet station is active; start / stop / reset buttons; speed or cycle time; OEE or shift totals; safety status (doors, light curtains). (4)

4.2 $\text{OEE} = \text{A} \times \text{P} \times \text{Q}$ ✓ $= 0{,}90 \times 0{,}95 \times 0{,}98$ ✓ $= 0{,}8379 = 83{,}79\%$ ✓ (3)

4.3 Any THREE ✓✓✓: managers can monitor the cell from a central control room; production and downtime data is stored for reports and trends; alarms are logged and sent to technicians; recipes / pallet patterns can be changed centrally; data helps to plan maintenance and improve OEE. (3)

4.4 Any TWO ✓✓: predictive maintenance using vibration / current monitoring; remote monitoring on phones or the cloud; automatic stock / order integration with ERP; AGVs to move pallets; vision to check box labels; digital twin for testing changes. (2)

4.5 Any TWO ✓✓: strong passwords and user levels on the HMI; firewalls / separate network for machines; keep software and firmware updated; disable unused ports; secure VPN for remote access; back up programs. (2)

4.6 Any ONE ✓✓: Profinet; EtherNet/IP; Modbus TCP; OPC UA; Profibus. (2)

[16]

QUESTION 5

5.1 Any FIVE ✓✓✓✓✓: perimeter fence; interlocked access gate; light curtains at the pallet openings; emergency-stop buttons inside and outside; safety PLC / relays; reduced speed and enabling device in teach mode; warning beacon; floor marking and signs; lock-out procedure; risk assessment to ISO 10218. (5)

5.2 Extra sensors detect the pallet (its size, timing and direction) entering the light curtain ✓; for that short moment the PLC "mutes" the curtain so the pallet can pass without stopping the cell ✓; a person does not create the same sensor pattern, so the curtain still stops the robot if a person tries to enter ✓. (3)

5.3 Any FOUR in a sensible order ✓✓✓✓: simulate the program and layout offline; check all wiring and test every input and output; test all safety functions (E-stops, gate interlocks, light curtains, muting); run the robot slowly in manual / teach mode through every point; do a dry run without boxes, then with boxes at low speed; increase to full speed and check the cycle time; train the operators. (4)

5.4 In automatic mode the robot can move suddenly, fast and with great force, so a person could be struck or crushed ✓✓. After a door has been opened the cell must not restart by itself: close the door, reset the safety circuit from outside the cell (after checking nobody is inside) and then press START ✓✓. (4)

[16]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE